

ENTHUSE

IIT MATHEMATICS

INVERSE TRIGONOMETRIC FUNCTION



EXERCISE # 1

Straight Objective Type

- The domain of the function $\sin^{-1}\left(\log_2\left(\frac{x}{3}\right)\right)$ is -
 (A) $\left[\frac{1}{2}, 3\right]$ (B) $\left(\frac{1}{2}, 3\right)$ (C) $\left[\frac{3}{2}, 6\right]$ (D) $\left[\frac{1}{2}, 2\right]$
- Domain of the function $f(x) = \log_e \cos^{-1}\{\sqrt{x}\}$ is, where $\{.\}$ represents fractional part function -
 (A) $x \in \mathbb{R}$ (B) $x \in [0, \infty)$ (C) $x \in (0, \infty)$ (D) $x \in \mathbb{R} - \{x \mid x \in I\}$
- $\tan^{-1}\left(1 - x^2 - \frac{1}{x^2}\right) + \sin^{-1}\left(x^2 + \frac{1}{x^2} - 1\right)$ (where $x \neq 0$) is equal to
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) π
- The value of $\tan^2(\sec^{-1}3) + \cot^2(\operatorname{cosec}^{-1}4)$ is -
 (A) 9 (B) 16 (C) 25 (D) 23
- If $x > 0$, $\cos^{-1}\left(\frac{12}{x}\right) = \frac{\pi}{2} - \cos^{-1}\left(\frac{16}{x}\right)$ then x is -
 (A) 12 (B) 16 (C) 20 (D) 320
- If $2 \leq a < 3$, then the value of $\cos^{-1} \cos[a] + \operatorname{cosec}^{-1} \operatorname{cosec}[a] + \cot^{-1} \cot[a]$, (where $[.]$ denotes greatest integer less than equal to x) is equal to
 (A) $2 - \pi$ (B) $2 + \pi$ (C) π (D) 6
- If $\cos^{-1}(2x^2 - 1) = 2\pi - 2\cos^{-1}x$, then -
 (A) $x \in [-1, 0]$ (B) $x \in [0, 1]$ (C) $x \in \left[0, \frac{1}{\sqrt{2}}\right]$ (D) $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$
- Number of integral ordered pairs (a, b) for which
 $\sin^{-1}(1 + b + b^2 + \dots \infty) + \cos^{-1}\left(a - \frac{a^2}{3} + \frac{a^3}{9} - \dots \infty\right) = \frac{\pi}{2}$ is -
 (A) 0 (B) 4 (C) 9 (D) Infinitely many
- $\lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1} \frac{2r+1}{r^4 + 2r^3 + r^2 + 1}$ is equal to -
 (A) $\frac{\pi}{4}$ (B) $\frac{3\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $-\frac{\pi}{8}$
- The range of the function $f(x) = \sin^{-1}(\log_2(-x^2 + 2x + 3))$ is -
 (A) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (B) $\left[-\frac{\pi}{2}, 0\right]$ (C) $\left[0, \frac{\pi}{2}\right]$ (D) $[-1, 1]$

11. Range of $f(x) = \cot^{-1}(\log_e(1 - x^2))$ is -
 (A) $(0, \pi)$ (B) $\left(0, \frac{\pi}{2}\right)$ (C) $\left[\frac{\pi}{2}, \pi\right)$ (D) $\left(0, \frac{\pi}{2}\right]$
12. The value of $\sin^{-1}\left\{\cot\left(\sin^{-1}\sqrt{\frac{2-\sqrt{3}}{4}} + \cos^{-1}\frac{\sqrt{12}}{4} + \sec^{-1}\sqrt{2}\right)\right\}$ is -
 (A) 5 (B) 6 (C) 0 (D) 10
13. Number of solution(s) of the equation $\cos^{-1}\sqrt{x} - \sin^{-1}\sqrt{x-1} + \cos^{-1}\sqrt{1-x} - \sin^{-1}\frac{1}{\sqrt{x}} = \frac{\pi}{2}$ is -
 (A) 0 (B) 1 (C) 2 (D) 4
14. $\sum_{r=0}^{\infty} \tan^{-1}\left(\frac{r((r+1)!)}{(r+1) + ((r+1)!)^2}\right)$ is equal to -
 (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{4}$ (C) $\cot^{-1}3$ (D) $\tan^{-1}2$
15. Let $f(x) = \begin{cases} x^2 - 4 & \text{if } |x| \leq 3 \\ 5\operatorname{sgn}|x-3| & \text{if } |x| > 3 \end{cases}$
 and $g(x) = 2\tan^{-1}(e^x) - \frac{\pi}{2}$ for all $x \in \mathbb{R}$, then which of the following is wrong?
 (where $\operatorname{sgn}(x)$ denotes signum function of x)
 (A) $f(x)$ is an even function (B) $gof(x)$ is an even function
 (C) $g(x)$ is an odd function (D) $fof(x)$ is an odd function

Linked Comprehension Type

Paragraph for Question 16 to 18

Consider a continuous function such that each image has atmost three pre image & atleast one image has exactly three pre images. This type of function is to be called as three-one function. On the basis of above information, answer the following questions :

16. Which of the following function is a three-one function ?
 (A) $|\ln|x||$ (B) $e^{|x|}$ (C) $x^3 + 3x^2 - 7x + 6$ (D) $\cos(\cos^{-1}x)$
17. If $f(x) = \sin^{-1}(\sin x)$ is a three-one function, then possible interval of x is -
 (A) $[-\pi, \pi]$ (B) $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$ (C) $(-2\pi, 0]$ (D) $\left[\frac{-3\pi}{2}, \frac{\pi}{2}\right]$
18. If $f(x)$ is a three-one function such that $f(a) = f(b)$ (where $a \neq b$), then number of maximum possible values of b is -
 (A) 1 (B) 2 (C) 3 (D) 4

EXERCISE # 2

1. (a) Find the following :

(i) $\tan \left[\cos^{-1} \frac{1}{2} + \tan^{-1} \left(\frac{-1}{\sqrt{3}} \right) \right]$

(ii) $\cos^{-1} \left(\cos \frac{7\pi}{6} \right)$

(iii) $\cos \left(\tan^{-1} \frac{3}{4} \right)$

(iv) $\tan \left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2} \right)$

(b) Find the following :

(i) $\sin \left[\frac{\pi}{2} - \sin^{-1} \left(\frac{-\sqrt{3}}{2} \right) \right]$

(ii) $\cos \left[\cos^{-1} \left(\frac{-\sqrt{3}}{2} \right) + \frac{\pi}{6} \right]$

(iii) $\tan^{-1} \left(\tan \frac{3\pi}{4} \right)$

(iv) $\sin \left(\frac{1}{4} \arcsin \frac{\sqrt{63}}{8} \right)$

2. Find the domain of definition the following functions.

(Read the symbols $[*]$ and $\{ * \}$ as greatest integers and fractional part functions respectively))

(i) $f(x) = \arccos \frac{2x}{1+x}$

(ii) $f(x) = \sqrt{\cos(\sin x)} + \sin^{-1} \frac{1+x^2}{2x}$

(iii) $f(x) = \sin^{-1} \left(\frac{x-3}{2} \right) - \log_{10}(4-x)$

(iv) $f(x) = \sin^{-1}(2x + x^2)$

(v) $f(x) = \frac{\sqrt{1-\sin x}}{\log_5(1-4x^2)} + \cos^{-1}(1-\{x\})$, where $\{x\}$ is the fractional part of x .

(vi) $f(x) = \sqrt{3-x} + \cos^{-1} \left(\frac{3-2x}{5} \right) + \log_6(2|x|-3) + \sin^{-1}(\log_2 x)$

(vii) $f(x) = e^{\sin^{-1} \left(\frac{x}{2} \right)} + \tan^{-1} \left[\frac{x}{2} - 1 \right] + \ln(\sqrt{x - [x]})$

3. Find the domain and range of the following functions.
(Read the symbols $[*]$ and $\{ * \}$ as greatest integers and fractional part function respectively)
- (i) $f(x) = \cot^{-1}(2x - x^2)$ (ii) $f(x) = \sec^{-1}(\log_3 \tan x + \log_{\tan x} 3)$
- (iii) $f(x) = \cos^{-1}\left(\frac{\sqrt{2x^2 + 1}}{x^2 + 1}\right)$ (iv) $f(x) = \tan^{-1}\left(\log_{\frac{4}{5}}(5x^2 - 8x + 4)\right)$
4. Identify the pair(s) of functions which are identical. Also plot the graphs in each case.
- (a) $y = \tan(\cos^{-1} x)$; $y = \frac{\sqrt{1-x^2}}{x}$ (b) $y = \tan(\cot^{-1} x)$; $y = \frac{1}{x}$
5. Let $y = \sin^{-1}(\sin 8) - \tan^{-1}(\tan 10) + \cos^{-1}(\cos 12) - \sec^{-1}(\sec 9) + \cot^{-1}(\cot 6) - \operatorname{cosec}^{-1}(\operatorname{cosec} 7)$.
If y simplifies to $a\pi + b$, then find $(a - b)$.
6. Show that : $\sin^{-1}\left(\sin \frac{33\pi}{7}\right) + \cos^{-1}\left(\cos \frac{46\pi}{7}\right) + \tan^{-1}\left(-\tan \frac{13\pi}{8}\right) + \cot^{-1}\left(\cot\left(-\frac{19\pi}{8}\right)\right) = \frac{13\pi}{7}$
7. Prove that : (a) $2\cos^{-1}\frac{3}{\sqrt{13}} + \cot^{-1}\frac{16}{63} + \frac{1}{2}\cos^{-1}\frac{7}{25} = \pi$ (b) $\arccos \sqrt{\frac{2}{3}} - \arccos \frac{\sqrt{6}+1}{2\sqrt{3}} = \frac{\pi}{6}$
8. If α and β are the roots of the equation $x^2 + 5x - 49 = 0$, then find the value of $\cot(\cot^{-1}\alpha + \cot^{-1}\beta)$.
9. If $a > b > c > 0$, then find the value of : $\cot^{-1}\left(\frac{1+ab}{a-b}\right) + \cot^{-1}\left(\frac{1+bc}{b-c}\right) + \cot^{-1}\left(\frac{1+ca}{c-a}\right)$.
10. Find all values of k for which there is a triangle whose angles have measure $\tan^{-1}\left(\frac{1}{2}\right)$, $\tan^{-1}\left(\frac{1}{2} + k\right)$ and $\tan^{-1}\left(\frac{1}{2} + 2k\right)$.
11. Find the simplest value of
- (a) $f(x) = \arccos x + \arccos\left(\frac{x}{2} + \frac{1}{2}\sqrt{3-3x^2}\right)$, $x \in \left(\frac{1}{2}, 1\right)$
- (b) $f(x) = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$, $x \in \mathbb{R} - \{0\}$

12. Prove that the identities :

(a) $\sin^{-1}\cos(\sin^{-1}x) + \cos^{-1}\sin(\cos^{-1}x) = \frac{\pi}{2}, |x| \leq 1$

(b) $\tan(\tan^{-1}x + \tan^{-1}y + \tan^{-1}z) = \cot(\cot^{-1}x + \cot^{-1}y + \cot^{-1}z)$

13. Solve the inequality : $(\text{arc sec } x)^2 - 6(\text{arc sec } x) + 8 > 0$

14. Solve the following :

(a) $\sin^{-1}x + \sin^{-1}2x = \frac{\pi}{3}$

(b) $\tan^{-1}\frac{1}{1+2x} + \tan^{-1}\frac{1}{1+4x} = \tan^{-1}\frac{2}{x^2}$

(c) $\tan^{-1}(x-1) + \tan^{-1}(x) + \tan^{-1}(x+1) = \tan^{-1}(3x)$

(d) $3\cos^{-1}x = \sin^{-1}(\sqrt{1-x^2}(4x^2-1))$

(e) $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3} \& \cos^{-1}x - \cos^{-1}y = \frac{\pi}{3}$

15. Find the sum of the series :

(a) $\cot^{-1}7 + \cot^{-1}13 + \cot^{-1}21 + \cot^{-1}31 + \dots$ to n terms.

(b) $\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{2}{9} + \dots + \tan^{-1}\frac{2^{n-1}}{1+2^{2n-1}} + \dots \infty$

(c) $\tan^{-1}\frac{1}{x^2+x+1} + \tan^{-1}\frac{1}{x^2+3x+3} + \tan^{-1}\frac{1}{x^2+5x+7} + \tan^{-1}\frac{1}{x^2+7x+13}$ to n terms

(d) $\sin^{-1}\frac{1}{\sqrt{5}} + \sin^{-1}\frac{1}{\sqrt{65}} + \sin^{-1}\frac{1}{\sqrt{325}} + \dots + \sin^{-1}\frac{1}{\sqrt{4n^4+1}} + \dots \infty$ terms.

EXERCISE # 3 (JM)

- If x, y, z are in A.P. and $\tan^{-1}x, \tan^{-1}y$ and $\tan^{-1}z$ are also in A.P., then [JEE (Main)-2013]

(1) $x = y = z$ (2) $2x = 3y = 6z$ (3) $6x = 3y = 2z$ (4) $6x = 4y = 3z$
- Let $\tan^{-1} y = \tan^{-1} x + \tan^{-1} \left(\frac{2x}{1-x^2} \right)$, where $|x| < \frac{1}{\sqrt{3}}$. Then a value of y is : [JEE (Main)-2015]

(1) $\frac{3x-x^3}{1+3x^2}$ (2) $\frac{3x+x^3}{1+3x^2}$ (3) $\frac{3x-x^3}{1-3x^2}$ (4) $\frac{3x+x^3}{1-3x^2}$
- If $f(x) = 2\tan^{-1}x + \sin^{-1} \left(\frac{2x}{1+x^2} \right)$, $x > 1$, then $f(5)$ is equal to [JEE (Main) Online-2015]

(1) $\pi/2$ (2) π (3) $4\tan^{-1}(5)$ (4) $\tan^{-1} \left(\frac{65}{156} \right)$
- A value of x satisfying the equation $\sin[\cot^{-1}(1+x)] = \cos[\tan^{-1}x]$, is [JEE (Main) Online-2017]

(1) $\frac{1}{2}$ (2) 0 (3) -1 (4) $-\frac{1}{2}$
- The value of $\tan^{-1} \left[\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \right]$, $|x| < \frac{1}{2}$, $x \neq 0$, is equal to [JEE (Main) Online-2017]

(1) $\frac{\pi}{4} - \frac{1}{2} \cos^{-1} x^2$ (2) $\frac{\pi}{4} + \frac{1}{2} \cos^{-1} x^2$ (3) $\frac{\pi}{4} - \cos^{-1} x^2$ (4) $\frac{\pi}{4} + \cos^{-1} x^2$
- If $\cos^{-1} \left(\frac{2}{3x} \right) + \cos^{-1} \left(\frac{3}{4x} \right) = \frac{\pi}{2}$ ($x > \frac{3}{4}$) then x is equal to : [JEE (Main) Online-2019]

(1) $\frac{\sqrt{145}}{12}$ (2) $\frac{\sqrt{145}}{10}$ (3) $\frac{\sqrt{146}}{12}$ (4) $\frac{\sqrt{145}}{11}$
- If $x = \sin^{-1}(\sin 10)$ and $y = \cos^{-1}(\cos 10)$, then $y - x$ is equal to [JEE (Main) Online-2019]

(1) 10 (2) 0 (3) π (4) 7π
- The value of $\cot \left(\sum_{n=1}^{19} \cot^{-1} \left(1 + \sum_{p=1}^n 2p \right) \right)$ is : [JEE (Main) Online-2019]

(1) $\frac{22}{23}$ (2) $\frac{23}{22}$ (3) $\frac{21}{19}$ (4) $\frac{19}{21}$

9. Considering only the principal values of inverse functions, the set

$$A = \left\{ x \geq 0 : \tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4} \right\}$$

- (1) is an empty set
(2) contains more than two elements
(3) contains two elements
(4) is a singleton

[JEE (Main) Online-2019]

10. If $\alpha = \cos^{-1}\left(\frac{3}{5}\right)$, $\beta = \tan^{-1}\left(\frac{1}{3}\right)$, where $0 < \alpha, \beta < \frac{\pi}{2}$, then $\alpha - \beta$ is equal to:

- (1) $\tan^{-1}\left(\frac{9}{14}\right)$ (2) $\cos^{-1}\left(\frac{9}{5\sqrt{10}}\right)$ (3) $\tan^{-1}\left(\frac{9}{5\sqrt{10}}\right)$ (4) $\sin^{-1}\left(\frac{9}{5\sqrt{10}}\right)$

[JEE (Main) Online-2019]

11. If $\cos^{-1}x - \cos^{-1}\frac{y}{2} = \alpha$, where $-1 \leq x \leq 1$, $-2 \leq y \leq 2$, $x \leq \frac{y}{2}$, then for all x, y , $4x^2 - 4xy \cos \alpha + y^2$ is equal to

- (1) $2 \sin^2 \alpha$ (2) $4 \cos^2 \alpha + 2x^2y^2$ (3) $4 \sin^2 \alpha$ (4) $4 \sin^2 \alpha - 2x^2y^2$

[JEE (Main) Online-2019]

12. The value of $\sin^{-1}\left(\frac{12}{13}\right) - \sin^{-1}\left(\frac{3}{5}\right)$ is

- (1) $\pi - \cos^{-1}\left(\frac{33}{65}\right)$ (2) $\pi - \sin^{-1}\left(\frac{63}{65}\right)$ (3) $\frac{\pi}{2} - \sin^{-1}\left(\frac{56}{65}\right)$ (4) $\frac{\pi}{2} - \cos^{-1}\left(\frac{9}{65}\right)$

[JEE (Main) Online-2019]

EXERCISE # 4 (JA)

1. The value of $\cot\left(\sum_{n=1}^{23} \cot^{-1}\left(1 + \sum_{k=1}^n 2k\right)\right)$ is [JEE(Advanced) 2013, 2]
- (A) $\frac{23}{25}$ (B) $\frac{25}{23}$ (C) $\frac{23}{24}$ (D) $\frac{24}{23}$

2. Match List-I with List-II and select the correct answer using the code given below the lists.

List-I

List-II

- P. $\left(\frac{1}{y^2} \left(\frac{\cos(\tan^{-1}y) + y\sin(\tan^{-1}y)}{\cot(\sin^{-1}y) + \tan(\sin^{-1}y)} \right)^2 + y^4 \right)^{1/2}$ takes value
- Q. If $\cos x + \cos y + \cos z = 0 = \sin x + \sin y + \sin z$, then possible value of $\cos \frac{x-y}{2}$ is
- R. If $\cos\left(\frac{\pi}{4} - x\right) \cos 2x + \sin x \sin 2x \sec x = \cos x \sin 2x \sec x + \cos\left(\frac{\pi}{4} + x\right) \cos 2x$, then possible value of $\sec x$ is
- S. If $\cot(\sin^{-1} \sqrt{1-x^2}) = \sin(\tan^{-1}(x\sqrt{6}))$, $x \neq 0$. Then possible value of x is

1. $\frac{1}{2}\sqrt{\frac{5}{3}}$

2. $\sqrt{2}$

3. $\frac{1}{2}$

4. 1

Codes :

	P	Q	R	S
(A)	4	3	1	2
(B)	4	3	2	1
(C)	3	4	2	1
(D)	3	4	1	2

[JEE(Advanced) 2013, 3, (-1)]

3. Let $f : [0, 4\pi] \rightarrow [0, \pi]$ be defined by $f(x) = \cos^{-1}(\cos x)$. The number of points $x \in [0, 4\pi]$ satisfying the equation $f(x) = \frac{10-x}{10}$ is [JEE(Advanced) 2014]

4. If $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3\cos^{-1}\left(\frac{4}{9}\right)$ where the inverse trigonometric functions take only the principal values, then the correct(s) is(are) [JEE 2015, 4M, -0M]
(A) $\cos\beta > 0$ (B) $\sin\beta < 0$ (C) $\cos(\alpha + \beta) > 0$ (D) $\cos\alpha < 0$

5. The number of real solutions of the equation [JEE(Advanced)2018, (3)-0]

$$\sin^{-1}\left(\sum_{i=1}^{\infty} x^{i+1} - x \sum_{i=1}^{\infty} \left(\frac{x}{2}\right)^i\right) = \frac{\pi}{2} - \cos^{-1}\left(\sum_{i=1}^{\infty} \left(-\frac{x}{2}\right)^i - \sum_{i=1}^{\infty} (-x)^i\right) \text{ lying in the interval } \left(-\frac{1}{2}, \frac{1}{2}\right)$$

is _____.

(Here, the inverse trigonometric functions $\sin^{-1}x$ and $\cos^{-1}x$ assume values in and $[0, \pi]$, respectively.)

6. Let $E_1 = \left\{x \in \mathbb{R} : x \neq 1 \text{ and } \frac{x}{x-1} > 0\right\}$ and $E_2 = \left\{x \in E_1 : \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right) \text{ is a real number}\right\}$

(Here, the inverse trigonometric function $\sin^{-1}x$ assumes value in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$).

Let $f : E_1 \rightarrow \mathbb{R}$ be the function defined by $f(x) = \log_e\left(\frac{x}{x-1}\right)$

and $g : E_2 \rightarrow \mathbb{R}$ be the function defined by $g(x) = \sin^{-1}\left(\log_e\left(\frac{x}{x-1}\right)\right)$.

List-I

- P. The range of f is
Q. The range of g contains
R. The domain of f contains
S. The domain of g is

List-II

1. $\left(-\infty, \frac{1}{1-e}\right] \cup \left[\frac{e}{e-1}, \infty\right)$
2. $(0, 1)$
3. $\left[-\frac{1}{2}, \frac{1}{2}\right]$
4. $(-\infty, 0) \cup (0, \infty)$
5. $\left(-\infty, \frac{e}{e-1}\right]$
6. $(-\infty, 0) \cup \left(\frac{1}{2}, \frac{e}{e-1}\right]$

The correct option is :

- (A) $P \rightarrow 4 ; Q \rightarrow 2 ; R \rightarrow 1 ; S \rightarrow 1$ (B) $P \rightarrow 3 ; Q \rightarrow 3 ; R \rightarrow 6 ; S \rightarrow 5$
(C) $P \rightarrow 4 ; Q \rightarrow 2 ; R \rightarrow 1 ; S \rightarrow 6$ (D) $P \rightarrow 4 ; Q \rightarrow 3 ; R \rightarrow 6 ; S \rightarrow 5$

[JEE(Advanced)2018, (3)-1]

7. For non-negative integer n , let

$$f(n) = \frac{\sum_{k=0}^n \sin\left(\frac{k+1}{n+2}\pi\right) \sin\left(\frac{k+2}{n+2}\pi\right)}{\sum_{k=0}^n \sin^2\left(\frac{k+1}{n+2}\pi\right)}$$

Assuming $\cos^{-1}x$ takes values in $[0, \pi]$, which of the following option is /are correct ?

[JEE(Advanced)-2019]

(1) $f(4) = \frac{\sqrt{3}}{2}$

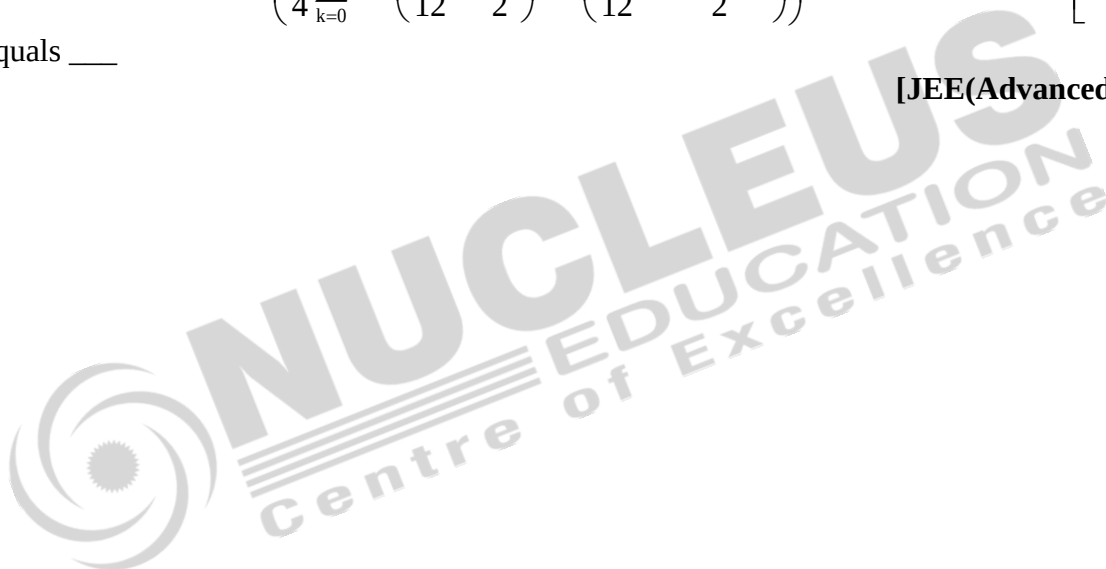
(2) $\lim_{n \rightarrow \infty} f(n) = \frac{1}{2}$

(3) $\sin(7\cos^{-1}f(5)) = 0$

(4) if $\alpha = \tan(\cos^{-1}f(6))$, then $\alpha^2 + 2\alpha - 1 = 0$

8. The value of $\sec^{-1}\left(\frac{1}{4}\sum_{k=0}^{10}\sec\left(\frac{7\pi}{12} + \frac{k\pi}{2}\right)\sec\left(\frac{7\pi}{12} + \frac{(k+1)\pi}{2}\right)\right)$ in the interval $\left[-\frac{\pi}{4}, \frac{3\pi}{4}\right]$ equals ____

[JEE(Advanced)-2019]



EXERCISE # 5

Multiple Correct Answer Type

- Let $f(x) = \sqrt{\cos^{-1}\sqrt{1-x^2} - \sin^{-1}x}$ then which of the following statement/s is/are correct -
 (A) Domain of $f(x)$ is $[-1,1]$ (B) Domain of $f(x)$ is $[0,1]$
 (C) Range of $f(x)$ is $\{0\}$ (D) Range of $f(x)$ is $[0, \sqrt{\pi}]$
- If $\alpha = 2\tan^{-1}\left(\sqrt{3-2\sqrt{2}}\right) + \sin^{-1}\left(\frac{1}{\sqrt{6}-\sqrt{2}}\right)$, $\beta = \cot^{-1}(\sqrt{3}-2) + \frac{1}{8}\sec^{-1}(-2)$ &
 $\gamma = \tan^{-1}\frac{1}{\sqrt{2}} + \cos^{-1}\frac{1}{\sqrt{3}}$, then
 (A) $\alpha = \beta$ (B) $\alpha + \beta = 3\gamma$ (C) $4(\beta - \gamma) = \alpha$ (D) $\beta = \gamma$
- If α is only real root of the equation $x^3 + (\cos 1)x^2 + (\sin 1)x + 1 = 0$, then $\left(\tan^{-1}\alpha + \tan^{-1}\frac{1}{\alpha}\right)$
cannot be equal to -
 (A) 0 (B) $\frac{\pi}{2}$ (C) $-\frac{\pi}{2}$ (D) π
- If $\sin\left(2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right) + \cos\left(2\tan^{-1}\left(\frac{1}{3}\right)\right) = \frac{p}{q}$, where p & q are relatively prime then digit at units
 place of $(p-q)^{2k+1}$, $k \in \mathbb{N}$, can be -
 (A) 1 (B) 3 (C) 7 (D) 9
- Consider the function $f(x) = e^x$ and $g(x) = \sin^{-1}x$, then which of the following is/are necessarily
 true.
 (A) Domain of $g \circ f$ = Domain of f (B) Range of $g \circ f \subset$ Range of g
 (C) Domain of $g \circ f$ is $(-\infty, 0]$ (D) Range of $g \circ f$ is $\left[-\frac{\pi}{2}, 0\right)$
- Let $f(x) = \sin^{-1}(\tan x) + \cos^{-1}(\cot x)$ then
 (A) $f(x) = \frac{\pi}{2}$ wherever defined (B) domain of $f(x)$ is $x = n\pi \pm \frac{\pi}{4}$, $n \in \mathbb{I}$
 (C) period of $f(x)$ is $\frac{\pi}{2}$ (D) $f(x)$ is many one function

7. Let $f(x) = e^{x^3 - x^2 + x}$ be an invertible function such that $f^{-1} = g$, then -
 (A) $g(e) = 0$ (B) Domain of 'g' is $\mathbb{R}^+$
 (C) Range of 'g' is $\mathbb{R}$ (D) $f(g(e)) = e$
8. Value of $3\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$ is greater than
 (A) $\frac{\pi}{2}$ (B) $\frac{2\pi}{3}$ (C) $\frac{3\pi}{4}$ (D) $\frac{5\pi}{6}$
9. Which of the following is/are correct ?
 (A) $\cot^{-1}(x) = \tan^{-1}\left(\frac{1}{x}\right) \forall x \in \mathbb{R} - \{0\}$
 (B) If $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = \text{sgn}(e^x)$ then $f(x)$ is an into function.
 (C) If $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $f(x) = \sin x + x$ then $f(x)$ is an odd function.
 (D) If $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = \frac{e^x}{e^{[x]}}$ then $f(x)$ is a periodic function.
 (where $[.]$ represents greatest integer function)

Subjective Type Questions

10. Least integral value of x for which inequality $\sin^{-1}\left(\sin\left(\frac{2e^x + 3}{e^x + 1}\right)\right) > \pi - \frac{5}{2}$ holds, is
11. Number of integral solutions of the equation $2\sin^{-1}\sqrt{x^2 - x + 1} + \cos^{-1}\sqrt{x^2 - x} = \frac{3\pi}{2}$ is
12. If $\sin^{-1}\sin\left(\frac{10\pi}{3}\right) + \cos^{-1}\cos\frac{22\pi}{3} + \tan^{-1}\tan 10 = a\pi + b$, then $\left|\frac{3ab}{80}\right|$ is equal to

EXERCISE # 6

1. If $\alpha = 2 \arctan\left(\frac{1+x}{1-x}\right)$ & $\beta = \arcsin\left(\frac{1-x^2}{1+x^2}\right)$ for $0 < x < 1$, then prove that $\alpha + \beta = \pi$, what the value of $\alpha + \beta$ will be if $x > 1$.
2. Solve the following :
 - (a) $\tan^{-1} \frac{x-1}{x+1} + \tan^{-1} \frac{2x-1}{2x+1} = \tan^{-1} \frac{23}{36}$
 - (b) $2\tan^{-1}x = \cos^{-1} \frac{1-a^2}{1+a^2} - \cos^{-1} \frac{1-b^2}{1+b^2}$ ($a > 0, b > 0$)
3. Find all the positive integral solutions of, $\tan^{-1}x + \cos^{-1} \frac{y}{\sqrt{1+y^2}} = \sin^{-1} \frac{3}{\sqrt{10}}$.
4. Prove that the equation, $(\sin^{-1}x)^3 + (\cos^{-1}x)^3 = \alpha\pi^3$ has no roots for $\alpha < \frac{1}{32}$ and $\alpha > \frac{7}{8}$.
5. Solve the following inequalities :
 - (a) $(\arccot x)^2 - 5 \arccot x + 6 > 0$
 - (b) $\arcsin x > \arccos x$
 - (c) $\tan^2(\arcsin x) > 1$
6. Solve the following system of inequalities :

$$4 \arctan^2 x - 8 \arctan x + 3 < 0 \quad \& \quad 4 \arccot x - \arccot^2 x - 3 \geq 0$$
7. Show that the roots r, s and t of the cubic $x(x-2)(3x-7) = 2$, are real and positive. Also compute the value of $\tan^{-1}(r) + \tan^{-1}(s) + \tan^{-1}(t)$.
8. Solve for x : $\sin^{-1}\left(\sin\left(\frac{2x^2+4}{1+x^2}\right)\right) < \pi - 3$.

ANSWER KEY

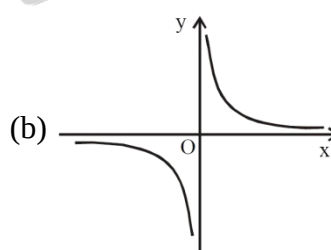
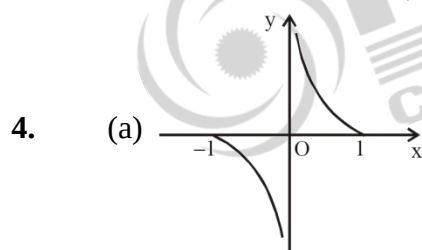
EXERCISE # 1

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. B | 3. B | 4. D | 5. C | 6. B | 7. A |
| 8. A | 9. A | 10. A | 11. C | 12. C | 13. A | 14. B |
| 15. D | 16. C | 17. A | 18. B | | | |

EXERCISE # 2

1. (a) (i) $\frac{1}{\sqrt{3}}$ (ii) $\frac{5\pi}{6}$, (iii) $\frac{4}{5}$, (iv) $\frac{17}{6}$; (b) (i) $\frac{1}{2}$, (ii) -1 , (iii) $-\frac{\pi}{4}$, (iv) $\frac{\sqrt{2}}{4}$
2. (i) $-1/3 \leq x \leq 1$ (ii) $\{1, -1\}$ (iii) $1 \leq x < 4$
 (iv) $[-(1+\sqrt{2}), (\sqrt{2}-1)]$ (v) $x \in (-1/2, 1/2), x \neq 0$ (vi) $(3/2, 2]$
 (vii) $(-2, 2) - \{-1, 0, 1\}$

3. (i) $D : x \in \mathbb{R}$ $R : [\pi/4, \pi)$
 (ii) $D : x \in \left(n\pi, n\pi + \frac{\pi}{2}\right) - \left\{x \mid x = n\pi + \frac{\pi}{4}\right\} n \in \mathbb{I}; R : \left[\frac{\pi}{3}, \frac{2\pi}{3}\right] - \left[\frac{\pi}{2}\right]$
 (iii) $D : x \in \mathbb{R}; R : \left[0, \frac{\pi}{2}\right)$
 (iv) $D : x \in \mathbb{R}; R : \left(-\frac{\pi}{2}, \frac{\pi}{4}\right]$



5. 53 6. 56 8. 10 9. π 10. $k = \frac{11}{4}$
11. (a) $\frac{\pi}{3}$; (b) $\frac{1}{2} \tan^{-1} x$ 13. $(-\infty, \sec 2) \cup [1, \infty)$
14. (a) $x = \frac{1}{2} \sqrt{\frac{3}{7}}$; (b) $x = 3$; (c) $x = 0, \frac{1}{2}, -\frac{1}{2}$;
 (d) $\left[\frac{\sqrt{3}}{2}, 1\right]$; (e) $x = \frac{1}{2}, y = 1$
15. (a) $\text{arc cot} \left[\frac{2n+5}{n} \right]$ (b) $\frac{\pi}{4}$, (c) $\text{arc tan}(x+n) - \text{arc tan} x$, (d) $\frac{\pi}{4}$

EXERCISE # 3 (JM)

1. 1 2. 3 3. 2 4. 4 5. 2 6. 1 7. 3
8. 3 9. 4 10. 4 11. 3 12. 3

EXERCISE # 4 (JA)

1. B 2. B 3. 3 4. B,C,D 5. 2.00 6. A
7. 1,3,4 8. 0.00

EXERCISE # 5

1. A,D 2. A,C 3. A,B,D 4. B,C 5. B,C 6. A,B,C,D
7. B,C,D 8. A,B 9. B,D 10. 1 11. 2 12. 1

EXERCISE # 6

1. $-\pi$ 2. (a) $x = \frac{4}{3}$; (b) $x = \frac{a-b}{1+ab}$; 3. $x = 1$; $y = 2$ & $x = 2$; $y = 7$
5. (a) $(\cot 2, \infty) \cup (-\infty, \cot 3)$ (b) $\left(\frac{\sqrt{2}}{2}, 1\right]$ (c) $\left(\frac{\sqrt{2}}{2}, 1\right) \cup \left(-1, -\frac{\sqrt{2}}{2}\right)$
6. $\left(\tan \frac{1}{2}, \cot 1\right]$ 7. $\frac{3\pi}{4}$ 8. $x \in (-1, 1)$